



The University of Azad Jammu & Kashmir, Muzaffarabad

Department of Software Engineering

Computer Networks

Lab No# 04

Task Title: Star Topology Using Switch

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Star Topology Using Switch

Introduction

A star topology is a network configuration in which all devices (computers, printers, etc.) are connected to a central device called a switch. The switch manages communication between devices, making the network efficient, reliable, and easy to troubleshoot.

Objective

To design and configure a star topology using a switch and assign IP addresses to ensure communication between devices.

Procedure

● Step 1: Open Simulation Software

Open Cisco Packet Tracer and prepare a blank workspace for network design.

● Step 2: Add Devices (Star Layout)

Place one switch at the center and add multiple PCs around it to form a star topology.

● Step 3: Connect Devices to Switch

Connect each PC to the switch using straight-through cables (all links should turn green).

● Step 4: Configure IP Addresses

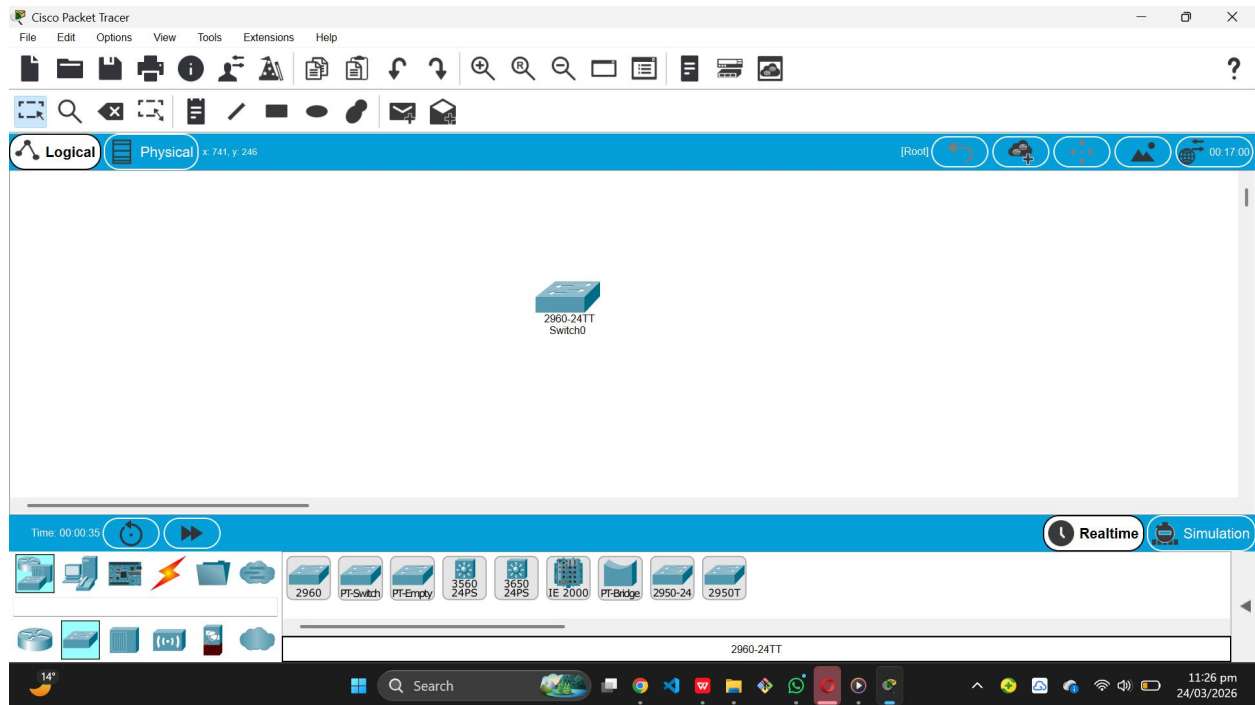
Assign unique IP addresses to each PC within the same network (e.g., 192.168.1.x with subnet mask 255.255.255.0).

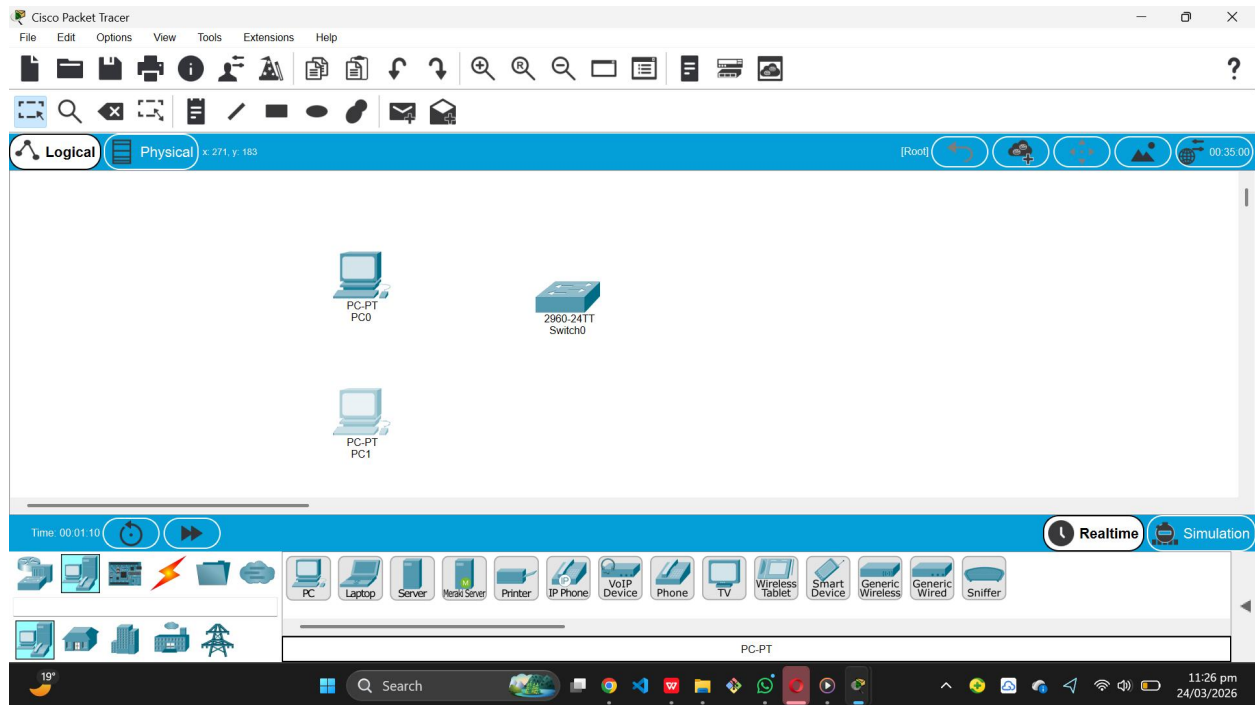
● Step 5: Verify Connectivity Using Ping and Packet Simulation

Use the ping command between PCs and also test using simulation mode to observe packet flow through the switch.

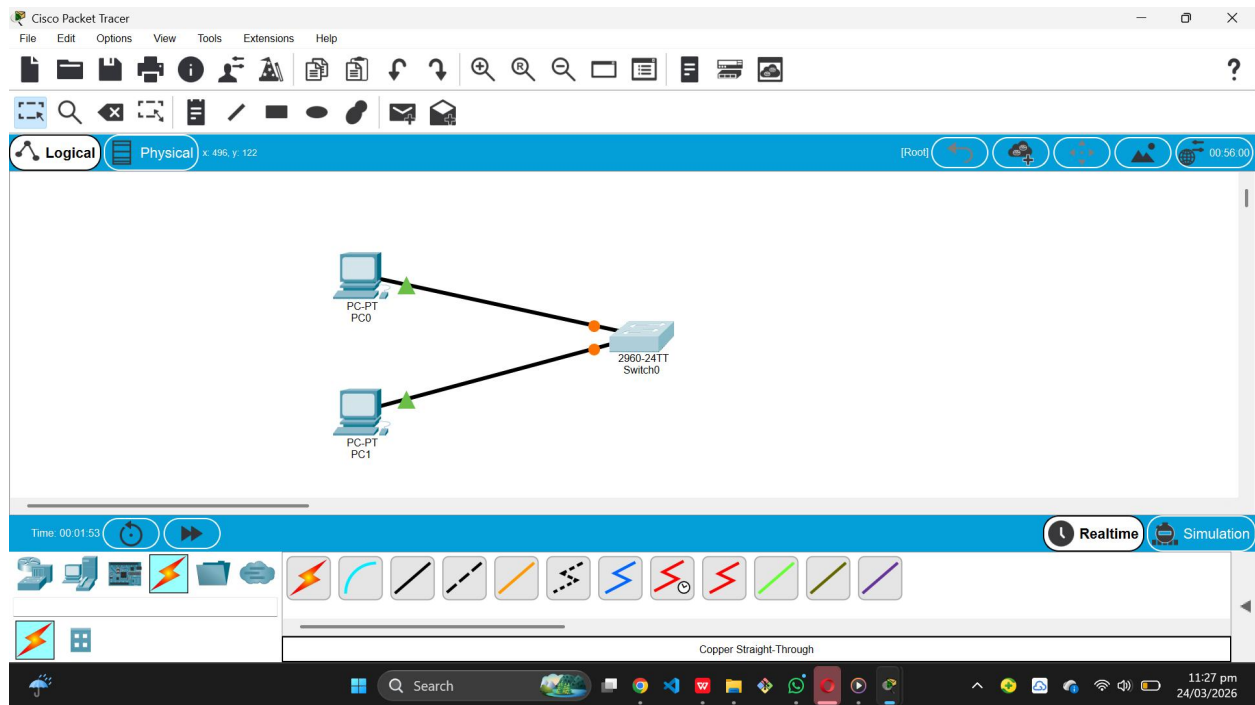
Screenshots

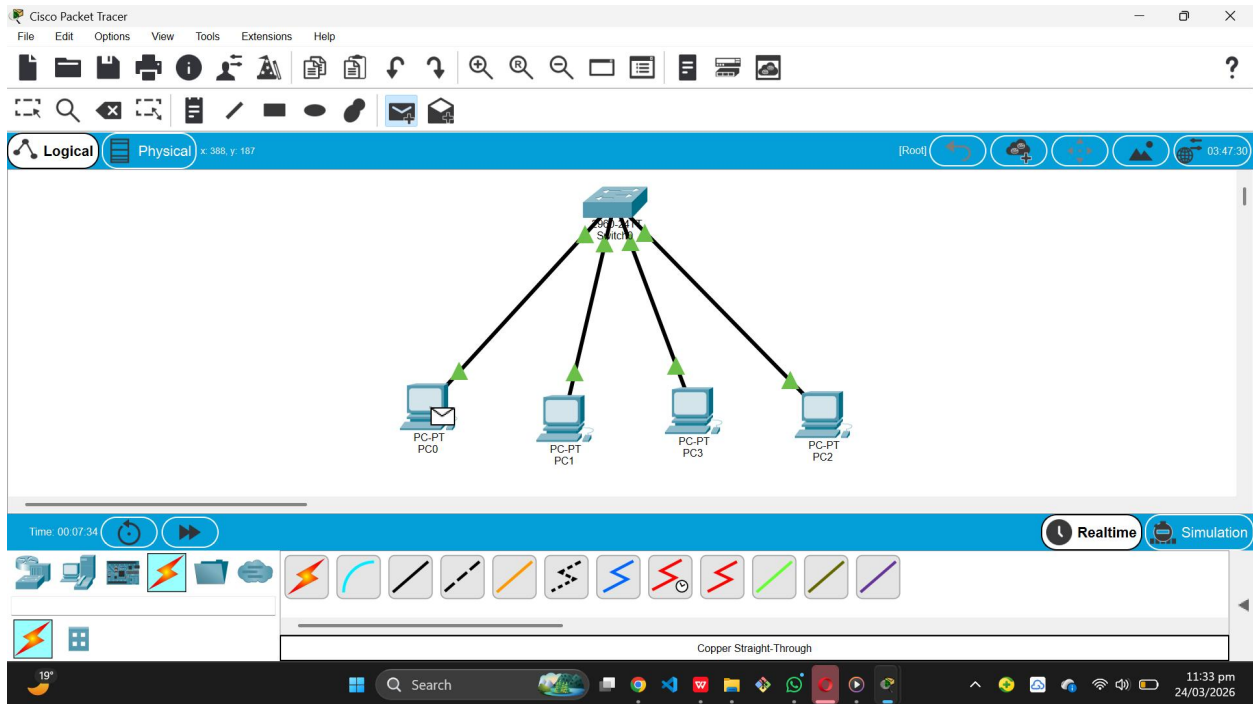
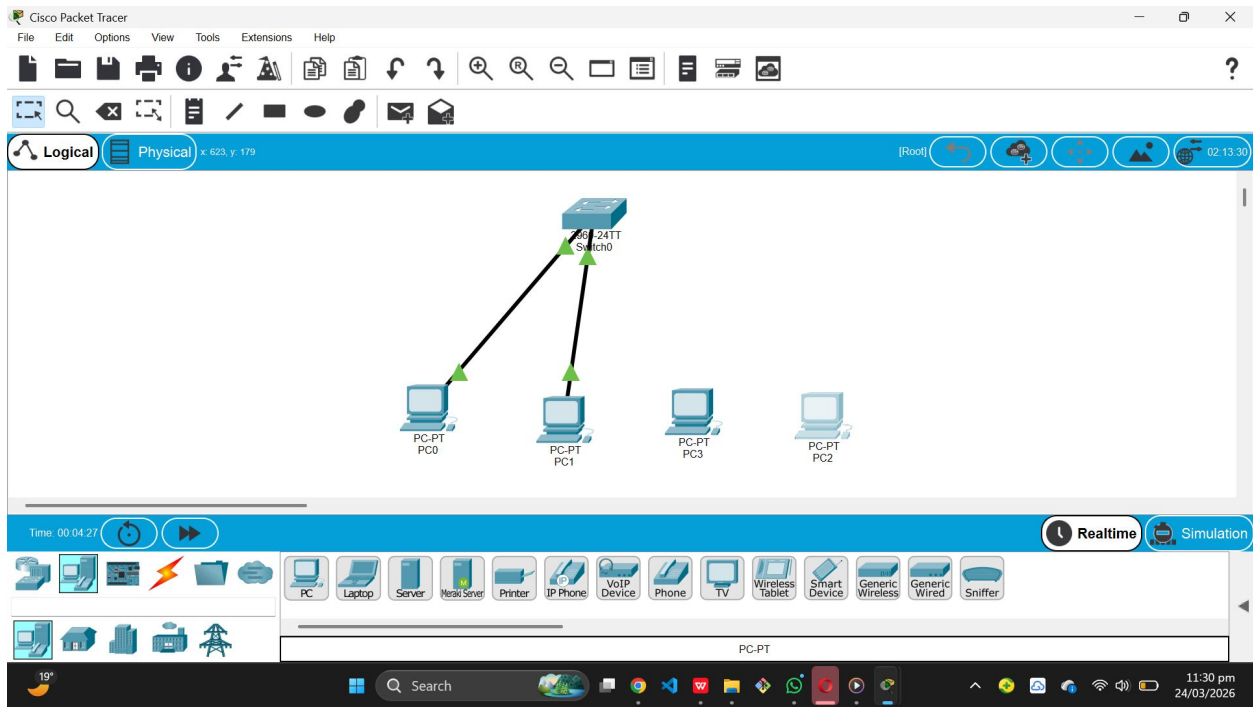
● Step 1 & 2: Adding Switch and PCs (Star Layout)



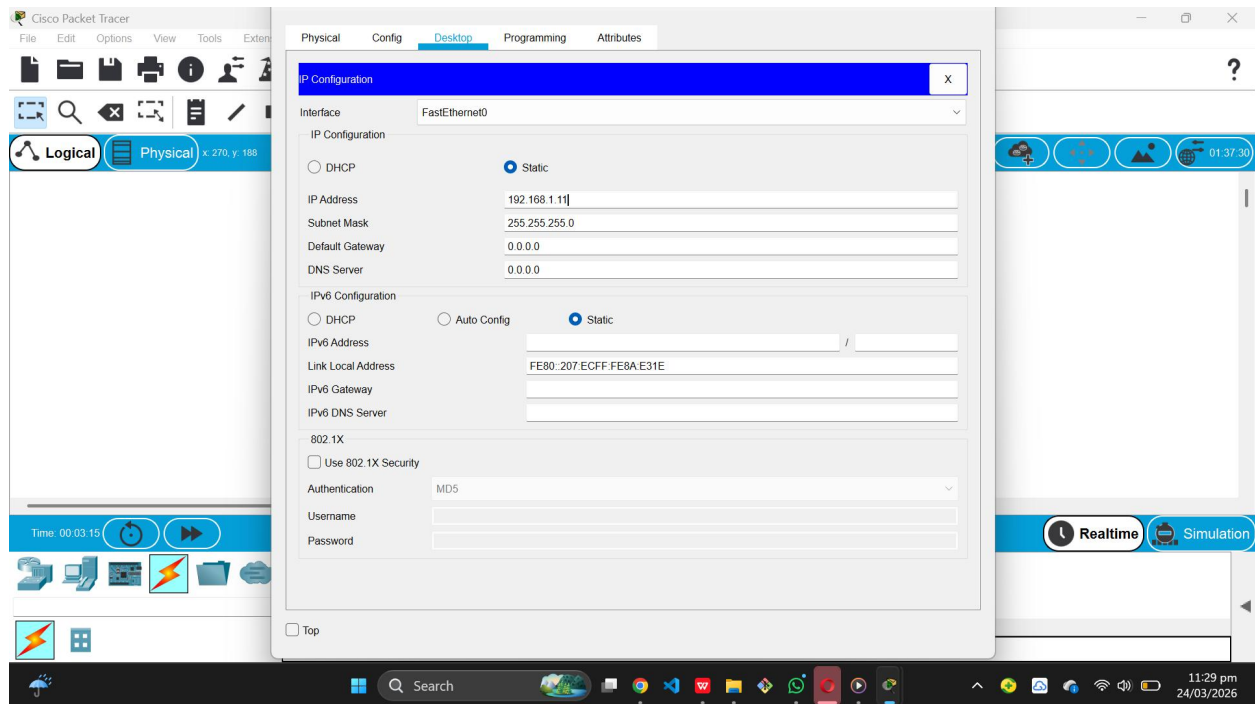
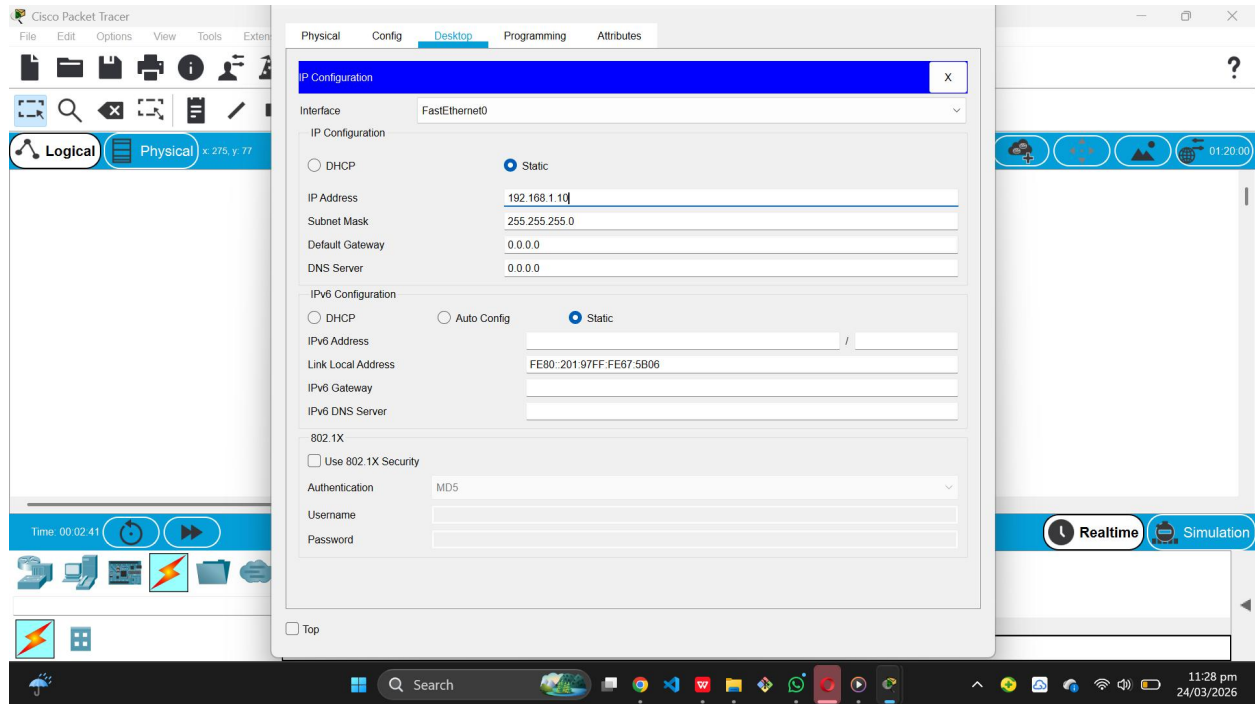


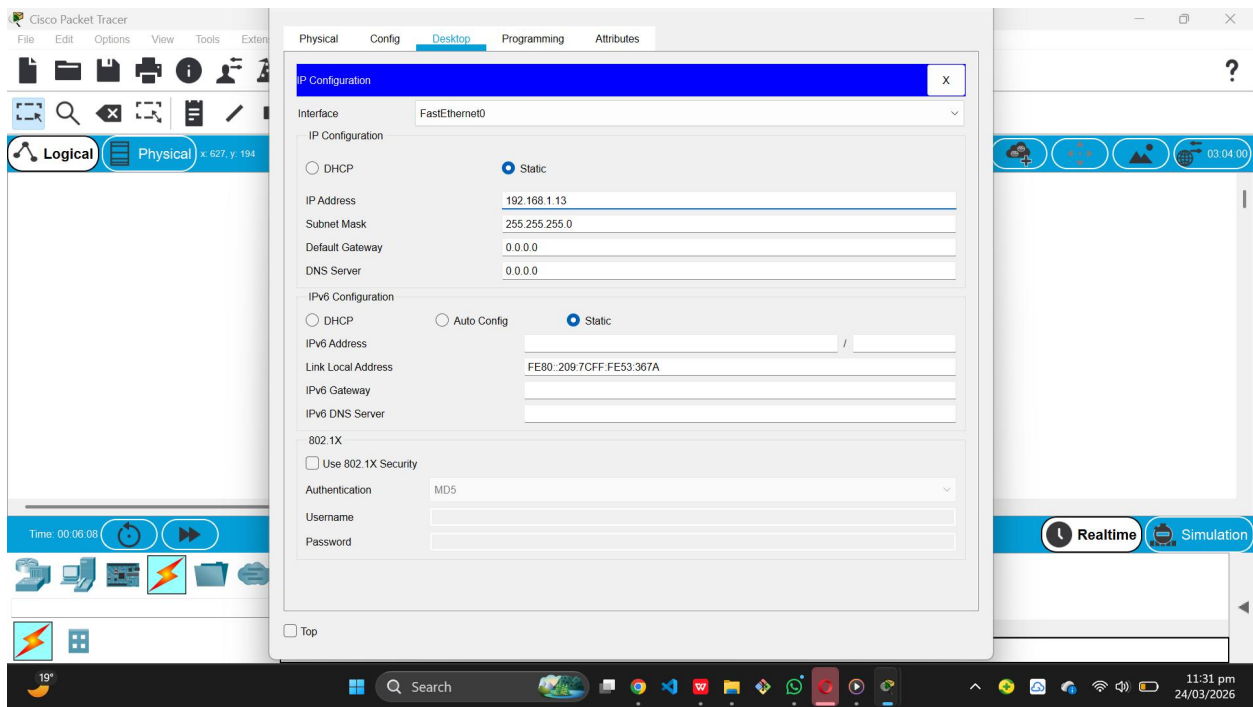
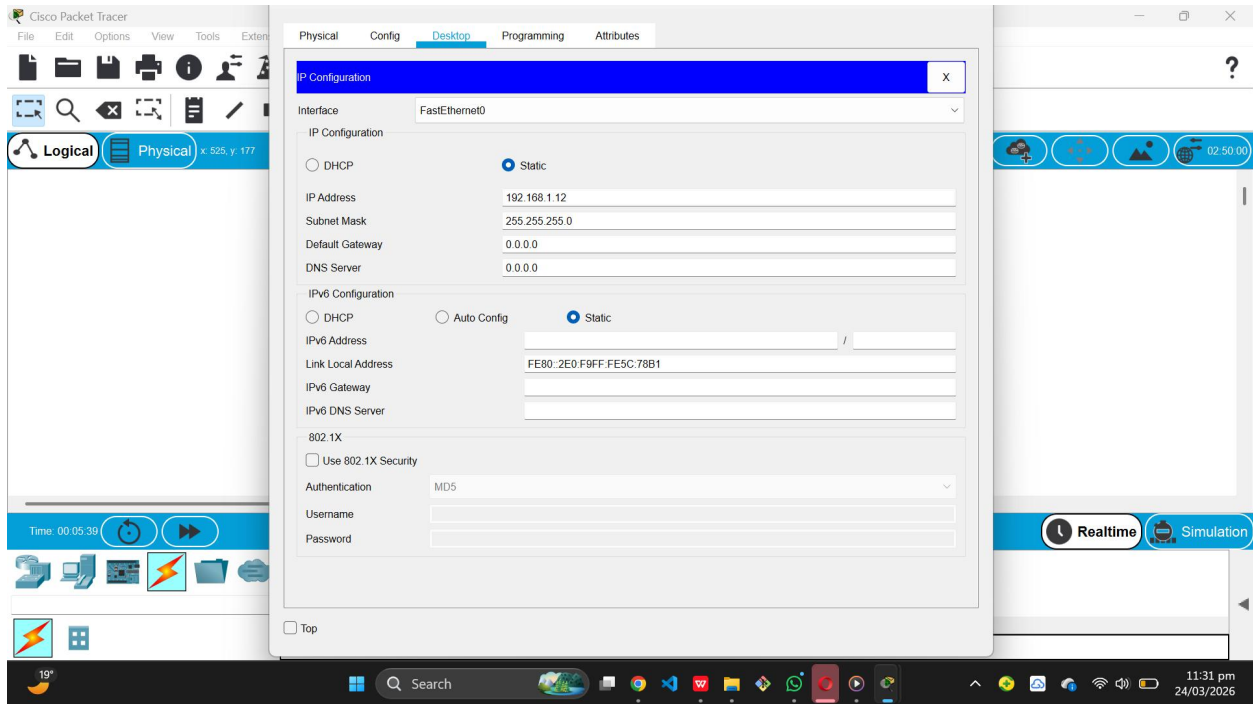
Step 3: Connecting Devices to Switch





● Step 4: Configuring IP Addresses (Multiple PCs)





● Step 5: Ping Results and Packet Simulation

```
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time=1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time=5ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 5ms, Average = 1ms
```

The screenshot displays the Cisco Packet Tracer interface. The main workspace shows a network topology with a central switch connected to four PCs (PC-PT PC0, PC-PT PC1, PC-PT PC2, and PC-PT PC3). The interface includes a menu bar (File, Edit, Options, View, Tools, Extensions, Help), a toolbar with various icons, and a status bar at the bottom. The 'Logical' tab is selected, showing the network diagram. The 'Simulation Panel' on the right displays an 'Event List' table with the following data:

Vis.	Time(sec)	Last Device	At Device	Type
1	0.000	PC0	PC0	ICMP

The 'PLAY CONTROLS' section at the bottom shows the simulation is running in 'Realtime' mode. The system clock at the bottom right indicates 11:33 pm on 24/03/2026.

Cisco Packet Tracer

File Edit Options View Tools Extensions Help

Logical Physical x: 407, y: 204

[Root] 04:16:00

Simulation Panel

Event List

Vis.	Time(sec)	Last Device	At Device	Type
	0.000	--	PC0	ICMP
	0.001	PC0	Switch0	ICMP
	0.002	Switch0	PC1	ICMP

Port: FastEthernet0 Link: Up IP Address: 192.168.1.11/24 IPv6 Address: <not set> MAC Address: 0007.EC8A.E31E
 Bluetooth: Down <not set> <not set> 000c.850c.4181

Gateway: <not set> DNS Server: <not set> Line Number: <not set> Physical Location: Intercity, Home City, Corporate Office

Time: 00:08:00.904 PLAY CONTROLS

Copper Straight-Through

11:34 pm 24/03/2026

Cisco Packet Tracer

File Edit Options View Tools Extensions Help

Logical Physical x: 370, y: 249

[Root] 04:49:30

Simulation Panel

Event List

Vis.	Time(sec)	Last Device	At Device	Type
	0.000	--	PC0	ICMP
	0.001	PC0	Switch0	ICMP
	0.002	Switch0	PC1	ICMP
	0.003	PC1	Switch0	ICMP
	0.004	Switch0	PC0	ICMP

Reset Simulation ☒ Constant Delay Capturing...

Time: 00:08:13.914 PLAY CONTROLS

Copper Straight-Through

11:35 pm 24/03/2026

Cisco Packet Tracer

File Edit Options View Tools Extensions Help

Logical Physical x: 597, y: 119 [Root] 05:13:30

Simulation Panel

Vis.	Time(sec)	Last Device	At Device	Type
	0.000	--	PC0	ICMP
	0.001	PC0	Switch0	ICMP
	0.002	Switch0	PC1	ICMP
	0.003	PC1	Switch0	ICMP
	0.004	Switch0	PC0	ICMP

Reset Simulation ☒ Constant Delay Captured to: 0.004 s

Time: 00:08:28.978 PLAY CONTROLS

Event List Realtime Simulation

Copper Straight-Through

19° Search 11:36 pm 24/03/2026

Cisco Packet Tracer

File Edit Options View Tools Extensions Help

Logical Physical x: 369, y: 246 [Root] 05:43:30

Simulation Panel

Vis.	Time(sec)	Last Device	At Device	Type
	0.000	--	PC0	ICMP
	0.001	PC0	Switch0	ICMP
	0.002	Switch0	PC1	ICMP
	0.003	PC1	Switch0	ICMP
	0.004	Switch0	PC0	ICMP
	0.345	--	Switch0	STP

Reset Simulation ☒ Constant Delay Captured to: 0.345 s

Time: 00:08:43.320 PLAY CONTROLS

Event List Realtime Simulation

Copper Straight-Through

19° Search 11:37 pm 24/03/2026

Conclusion

The star topology using a switch provides efficient communication and centralized control. It is easy to manage and troubleshoot, making it one of the most commonly used network topologies.